Exercise 5

Exercise 5 — Income statement 2017 with discounts, payroll, depreciation, taxes and dividends

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Prepare the income statement of a company with the following information for 2017:

- *Gross sales during the year reached 2,002 million €. The company's trade policy gives a 3% discount to its Type A customers, those whose annual purchases from the company exceed 120 million €. Type A customers account for 40% of sales. Assume total sales = gross sales − discount.*
- *Purchases of raw materials amounted to 962 million €.*
- *The workforce consists of 7,800 employees. The average salary is 30,000 € per year, and the average spending on Social Security is 30% of the salary.*
- *Payments for other operating expenses amounted to 457 million €.*
- *In recent years the company has invested a total of 1,200 million € in fixed assets with an average lifespan of 10 years (linear depreciation, no residual value).*
- *The company obtained an international loan of 225 million € in 2015, at 6% interest along the whole year; the whole principal will be repaid in 2020.*
- *The corporate tax rate is 30%. The company has agreed to a dividend of 50%.*
- *Assume no change in stocks of raw materials or finished products during the year.*

Phase 1 — Setup: units and hypotheses

All amounts are expressed in **millions of euros (M€)**. Salaries are given per employee in €, so they must be converted: $1\ \text{M€} = 10^6\ €$.

Hypotheses:

1. "No change in stocks" means initial inventory = final inventory, so $\text{COGS} = \text{Purchases}$.
2. The 120 M€ threshold only defines who is a Type A customer; the statement already says these customers make 40% of the sales, so the discount applies to that 40%.
3. The loan principal is not an expense: only its interest enters the income statement [Slide 11].
4. The dividend of 50% is applied to the **net income** [Slide 23].

Phase 2 — Frames: line of each item

Item	Line
Gross sales − discount	Sales revenue (net)
Raw materials (no stock change)	Cost of goods sold
Salaries + Social Security	Operating expense
Other operating expenses	Operating expense
Depreciation of fixed assets	Operating expense [Slide 10]
Interest on the loan	Non-operating item [Slide 11]
Tax 30%	Income taxes
Dividend 50%	Distribution of net income (not an expense)

Phase 3 — Derivation

Step 1. Net sales

$$\text{Type A sales} = 0.40 \times 2{,}002 = 800.8 \text{ M€}, \qquad \text{Discount} = 0.03 \times 800.8 = 24.024 \text{ M€}$$

$$\text{Net sales} = 2{,}002 - 24.024 = 1{,}977.976 \text{ M€}$$

Step 2. Personnel expense

$$\text{Salaries} = 7{,}800 \times 30{,}000 = 234{,}000{,}000 \text{ €} = 234 \text{ M€}$$

$$\text{Social Security} = 0.30 \times 234 = 70.2 \text{ M€}, \qquad \text{Personnel} = 234 + 70.2 = 304.2 \text{ M€}$$

Step 3. Depreciation and interest

$$D = \frac{1{,}200 - 0}{10} = 120 \text{ M€/year}, \qquad I = 225 \times 0.06 = 13.5 \text{ M€}$$

Step 4. Income statement 2017

Line	Computation	M€
Net sales	Step 1	1,977.976
– Cost of goods sold (raw materials)		962.000
= Gross profit	$1{,}977.976 - 962$	**1,015.976**
– Personnel	Step 2	304.200
– Other operating expenses		457.000
– Depreciation	Step 3	120.000
= Income from operations	$1{,}015.976 - 881.2$	**134.776**
– Interest expense	Step 3	13.500
= Income before taxes	$134.776 - 13.5$	**121.276**
– Income taxes (30%)	0.30×121.276	36.383
= Net income	$121.276 - 36.3828$	**84.893**

Step 5. Distribution of the net income [Slide 23]

$$\text{Dividends} = 0.50 \times 84.8932 = 42.447 \text{ M€}, \qquad \text{Retained earnings} = 84.8932 - 42.4466 = 42.447 \text{ M€}$$

Pay-out 50%, self-financing 50%.

Phase 4 — Interpretation and checks

- **Applying the discount to all the sales** would deduct $0.03 \times 2{,}002 = 60.06$ M€ instead of 24.024: the 40% restriction matters.
- **The loan repayment in 2020** does not appear in 2017: repaying principal is a payment that is never an expense.
- **Exact values:** income taxes 36.3828 M€ and net income 84.8932 M€ (shown rounded to three decimals in the table).
- **Dividends** come after taxes and are not an expense: the income statement ends at the net income, and the 50/50 split is shown separately.

Check against the official key

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Every line coincides with the key: discount 24.024; net sales 1,977.976; cost of goods sold 962; personnel 304.2; other operating expenses 457; depreciation 120; EBIT 134.776; interest 13.5; income before taxes 121.276; tax 36.3828; net income 84.8932; dividends 42.4466 and retained earnings 42.4466 (all in M€).